

Additional Material: Queuing systems

This section about queuing system is provided as optional reading for students interested in applications of stochastic processes and simulation. It is not part of the syllabus.

Queuing systems appear in many different areas. E.g. customers waiting to be served in a bank or post office, phone queues, patients waiting to be served by a doctor or hospital, boats waiting to be served by a harbour, computer programs waiting for server time, failed equipment waiting to be repaired, etc, etc.

Queuing systems are characterised by the arrival process (how “customers” arrive), the serving process (how customers are served) and the number of servers.

A particular notation has evolved in the queuing theory literature for describing these mechanisms for “first come first serve” queues:

arrival	/	service time	/	number of
process		distribution		servers

Some examples:

- $M/M/c$: Arrival as Poisson process, exponentially distributed service times, c servers. (E.g. a bank)
- $M/M/\infty$: Arrival as Poisson process, exponentially distributed service times, infinitely many servers. (E.g. an internet bank)
- $GI/M/\infty$: General independent interarrival times, exponentially distributed service times, ∞ servers.
- $M/G/1$: Arrival as Poisson process, general distribution of service times, one server. (E.g. an office)
- $D/M/c$: Deterministic interarrival times, exponentially distributed service times, c servers.
- $M/D/c$: Arrival as Poisson process, deterministic service times, c servers.

Studying queue systems by simulations

We shall consider $M/G/\infty$ queues, and in particular use simulations to study the number of customers in the system at different times under different scenarios. This is important for considerations of dimensioning to be certain that a system has sufficient capacity.

E.g., how many customers do an internet bank solution have to be able to handle at any time?

The number of customers in the system over time can be studied by the following procedure:

1. Simulate arrival times to the system, $a_1, a_2, \dots$, e.g. by a nonhomogeneous Poisson process or RP.
2. Simulate a service time, c_i , $i = 1, 2, \dots$, for each of the arrivals from a relevant service time distribution.
3. Calculate the departure times $d_1 = a_1 + c_1$, $d_2 = a_2 + c_2, \dots$
4. Sort the arrival and the departure times.

5. Run through the arrival and departure times and record the time and the number of customers in the system at any time point where there is a new arrival or departure.
6. Return the time points and the numbers of customers at each time point calculated in point 5.
7. Repeat 1-6 many times. Make plots or relevant summary statistics.

Example:

Customers logging in to an internet bank solution over the day is described by a non-homogeneous Poisson process with intensity:

$$\lambda(t) = 25 + 50t \sin(t \cdot 2\pi/24)^2 + 250 \sin(t \cdot \pi/24)^2$$

The amount of time the customers spend in the bank is described by a gamma distribution with parameters $\alpha = 2$ and $\beta = 0.1$.

How many customers will there be in the internet bank at different times during the day?

See R code: `stochastic_process_examples.R`